article returns to island ℓ_z). Before the decrease in homophily (where $p_d < p'_d$), we can construct an isomorphic diffusion process where an article remains within the same island (instead of switching to a different one) with probability $(p'_d - p_d)/p_d$. By construction of the cycle, whenever such an event occurs, the cycle becomes complete and the islands reached thereafter in the (p'_s, p'_d) sharing network are not (for that given cycle). Measuring across all cycles that occur in the (p'_s, p'_d) model, (weakly) more islands are reached than under the more homophilic (p_s, p_d) model. Consequently, virality is higher under the (p'_s, p'_d) sharing network (with less homophily) than with the (p_s, p_d) sharing network (more homophily). This establishes part (b). ∎

Proof of Proposition 3. Let us define r^* as

$$r^* \equiv \phi^{-1}\left(\max\left\{\frac{(1-q)(\tilde{u}-\tilde{c})}{(p-q)(\tilde{u}-\tilde{c})+(1-p)\tilde{u}}, \frac{c}{u+c}\right\}\right) \in (0,1)\,.$$

For part (a), first consider the case of $r < r^*$ and $p_d = 0$ (by continuity, the result extends to the case of sufficiently large p_s/p_d). In the most-sharing equilibrium, the seed agent most conducive to the article's spread is on the right-wing island, and given that $p_d = 0$, the equilibrium on the left-wing island is immaterial to its virality. Let us denote the right-wing island cutoffs by (b^*_R, b^{**}_R) (cutoffs are only island-dependent, per Lemma 1). Similar to the proof of Theorem 2(a), we define a cutoff space $\mathbf{B}_R$ such that $(\hat{b}^*_R, \hat{b}^{**}_R) \in \mathbf{B}_R$ if and only if $(\hat{b}^*_R, \hat{b}^{**}_R) \preceq (b^*_R, b^{**}_R)$. Similarly, we define the map $\varphi : \mathbf{B}_R \to \mathbf{B}_R$ which maps an arbitrary cutoff $(\hat{b}^*_R, \hat{b}^{**}_R)$ to best-response cutoffs $(\hat{b}^{*,BR}_R, \hat{b}^{**,BR}_R)$. To show the map is well-defined, consider $U_R(\mathcal{S})$ *before* the increase in divisiveness or polarization and $U'_R(\mathcal{S})$ *after* the increase in divisiveness or polarization. Because the network structure is fixed, note that $U^{(2)}_R(\mathcal{S}) = U^{(2)'}_R(\mathcal{S})$ when the cutoffs $(\hat{b}^*_R, \hat{b}^{**}_R)$ are taken as given, so the difference $U'_R(\mathcal{S}) - U_R(\mathcal{S})$ depends only on the difference between $U^{(1)}_R(\mathcal{S})$ and $U^{(1)'}_R(\mathcal{S})$. Specifically, the difference in share payoff depends only on the change in π_i following the increase in divisiveness or polarization. Observe that:

$$\begin{aligned}
\frac{\partial \pi_i}{\partial p} &= \frac{(2b_i-1)(1-\phi(r))\phi(r)(1-q-b_i(1-2q))}{(b_i(2p\phi(r)+2q(1-\phi(r))-1)-p\phi(r)-q(1-\phi(r))+1)^2} > 0\,;\\
\frac{\partial \pi_i}{\partial q} &= \frac{(2b_i-1)(1-\phi(r))\phi(r)(1-p+b_i(2p-1))}{((2b_i-1)\phi(r)(p-q)+2b_iq-b_i-q+1)^2} > 0\,,
\end{aligned}$$

whenever $b_i > 1/2$. Likewise, as we showed in Lemma A.1, $\partial\pi_i/\partial b_i > 0$ for all b_i and greater polarization increases ideological priors for agents with $b_i > 1/2$ (by Lemma A.3). By virtue of $\underline{b}_R > 1/2$, we observe that $U^{(1)'}_R(\mathcal{S}) > U^{(1)}_R(\mathcal{S})$, and so $U'_R(\mathcal{S}) > U_R(\mathcal{S})$. Thus, as in the proof of Theorem 2(a), φ is well-defined. Applying the Tarski fixed-point theorem (see Tarski (1955)), we find that the most-sharing equilibrium leads to more sharing in the right-wing island. Because the network structure $\mathbf{P}$ remains constant and there is a uniform shift in sharing, our weaker notion of virality also increases.

For part (b), consider $r \geq r^*$. Note that for $r \geq r^*$, ignoring is a better response to disliking for any agent (regardless of the agent's prior) and sharing is a better response to ignoring for all $b_i > 1/2$. The former follows from noting $\pi_i \geq \frac{\tilde{u}-\tilde{c}}{\tilde{u}}$ for an agent with prior $b_i = 0$ and the latter from noting $\pi_i \geq \frac{c}{u+c}$